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Initial Post

by [Milosz Gembicki](#) - Sunday, 17 May 2026, 2:10 PM

Kovaitė and Stankevičienė (2019) use Industry 4.0 to describe the digital transformation of business models through connected, data-driven and automated technologies. In simple terms, it is the “fourth industrial revolution”, where traditional processes are changed by technologies such as the Internet of Things, cloud computing, automation, artificial intelligence and cyber-physical systems. For example, a manufacturing firm may use IoT sensors to monitor machinery in real time, while an energy company may use smart meters and digital platforms to manage customer consumption and billing more efficiently.

The authors argue that digitalisation creates opportunity but also introduces new risks across six categories: technical, competence, staff acceptance, customer/partner acceptance, data privacy and security, and financial risk (Kovaitė and Stankevičienė, 2019). One real-world example is the Colonial Pipeline ransomware attack, where a cyberattack led to a six-day shutdown and affected fuel supply, showing technical, security and financial risk in a highly digitalised energy system. Another example is the British Airways data breach, where customer data was compromised and the ICO initially announced its intention to fine BA £183.39 million under GDPR, demonstrating data privacy, security and reputational risk.

Ghobakhloo (2020) supports Kovaitė and Stankevičienė’s argument by showing that Industry 4.0 can improve sustainability, efficiency and business model innovation, but requires careful implementation because its benefits depend on the relationship between technologies, processes and organisational readiness. Therefore, digitalisation should not be treated as a purely technical upgrade. It should be assessed as a risk management process where threats, vulnerabilities and assets are considered together. In Open FAIR terms, this could be expressed as: $\text{Risk} = \text{Threat} \times \text{Vulnerability} \times \text{Asset}$.

References

Ghobakhloo, M. (2020) ‘Industry 4.0, digitization, and opportunities for sustainability’, *Journal of Cleaner Production*, 252, 119869.

Kovaitė, K. and Stankevičienė, J. (2019) ‘Risks of digitalisation of business models’, *Contemporary Issues in Business, Management and Economics Engineering*, pp. 379–388.

Tsvetanov, T. (2021) ‘The effect of the Colonial Pipeline shutdown on gasoline prices’, *Economics Letters*, 209, 110122.

European Data Protection Board (2019) ‘ICO statement: Intention to fine British Airways £183.39m under GDPR for data breach’.

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Re: Initial Post

by [Lauren Pechey](#) - Sunday, 24 May 2026, 12:49 PM

Hi Milosz,

Your discussion response is well-structured and provides a clear explanation of Industry 4.0 and its associated risks. You effectively summarised the key points made by Kovaitė and Stankevičienė and supported your discussion with relevant real-world examples. The inclusion of the Colonial Pipeline ransomware attack and the British Airways data breach was particularly effective because both cases clearly demonstrate how digital transformation can expose organisations to cybersecurity, operational, financial, and reputational risks.

I also found your use of Ghobakhloo (2020) highly relevant, as it strengthens the argument that In Chat to us! viewed not only as a technological advancement but also as a strategic and risk management challenge. Your organisations must consider threats, vulnerabilities, and assets together aligns strongly with information security frameworks and demonstrates critical understanding of risk assessment principles. Additionally, the inclusion of the Open FAIR risk equation added a strong

analytical perspective to your discussion and linked the topic effectively to cybersecurity risk management methodologies.

One further point that could strengthen your discussion is mentioning that Industry 4.0 environments often increase the attack surface due to interconnected devices and cloud-based systems, making continuous monitoring and employee cybersecurity awareness essential controls (Xu, Xu and Li, 2018). Overall, your response demonstrates strong research, critical analysis, and effective application of real-world examples to support the discussion topic.

References

Ghobakhloo, M. (2020) 'Industry 4.0, digitization, and opportunities for sustainability', *Journal of Cleaner Production*, 252, 119869.

Kovaitė, K. and Stankevičienė, J. (2019) 'Risks of digitalisation of business models during industry 4.0', *Journal of Business Economics and Management*, 20(6), pp. 1079–1098.

Xu, L.D., Xu, E.L. and Li, L. (2018) 'Industry 4.0: state of the art and future trends', *International Journal of Production Research*, 56(8), pp. 2941–2962.

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